

WestQuant Hamiltonian Representation Explorer (Project B)

Slide 1 — Project Overview

Problem: A mathematical optimization problem does not determine a unique useful Hamiltonian representation. The penalty strength, variable encoding, and landscape all matter.

Approach: Generate multiple mathematically valid Hamiltonian representations (scaling, permutation, bit-complement, MWIS penalty family) and verify equivalence by exhaustive state-by-state comparison. Each is classified: EXACT_EQUIVALENT, GROUND_STATE_EQUIVALENT, SAME_PROBLEM_DIFFERENT_DYNAMICS, APPROXIMATE, or INVALID.

Results: For MWIS, penalty strengths U in $[5.5, 22.2]$ all yield GROUND_STATE_EQUIVALENT Hamiltonians. The Hamiltonian main effect is $\eta^2(H) = 6.8\%$ (median across 6 problems).

Importance: The Hamiltonian representation choice matters less than its interaction with the embedding. Representation itself is an optimization variable, but the $H \times R$ interaction (48.2%) is the dominant effect.

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Slide 2 — QoolQit Experience

Positive aspects: QoolQit's `DataGraph.from_matrix` and `interaction_matrix()` made it easy to connect logical Hamiltonians to physical interactions. The $1/r^6$ Rydberg convention is natural for QUBO/MWIS. `Register.from_coordinates` enables rapid coordinate experimentation.

Challenges: Native Rydberg interactions are repulsive ($J > 0$), so Hamiltonians with negative pair interactions (e.g., after bit-complement) are not directly native-realizable. The physical realizability classifier must flag this before expensive embedding. DMM is needed for MWIS node weights.

Workflow: The five-class equivalence taxonomy forced mathematical rigor. Exhaustive verification for $n \leq 16$ ensured no false equivalence claims. The realizability classifier prevents wasted computation on non-native Hamiltonians.